

SYNTACTIC VS. MORPHOLOGICAL VERBAL CONCORD ACROSS AUSTRONESIAN

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1 GOALS

- Investigate verbal concord and its implications for morphosyntactic theory
- Compare voice and ϕ -morphology in Serial Verb Constructions (SVCs) across Austronesian, spanning the Oceanic, Formosan, and Timoric sub-branches
- Propose a distinction between syntactic and morphological verbal concord
- Propose an account of syntactic concord under an interaction/satisfaction (INT/SAT) approach to Agree ([Deal 2015, 2021](#)) via the mechanism of goal flagging
- Present a typology of verbal concord patterns attested cross-linguistically

Roadmap: §2 SVCs · §3 Syntactic concord · §4 Morphological concord · §5 Alternatives · §6 Typology · §7 Conclusion

2 WHAT ARE SERIAL VERB CONSTRUCTIONS (SVCs)?

- Descriptively, SVCs typically include the following properties:
 - ▷ Two or more contiguous verbs
 - ▷ No overt coordination or subordination
 - ▷ A single prosodic domain, similar to other monoclausal constructions
 - ▷ Shared tense, aspect, mood (TAM) and polarity marking
 - ▷ A single predicate or “event”
- (1) a. *ó tí-wà-rà éteré à*
 he hit-split.open-TENSE plate the
 ‘He shattered the plate.’ [Igbo, Niger-Congo; [Lord 1975](#)]
- b. *yô:h bə-hām-yɔw*
 medicine spill-go-happen.straight.away
 ‘The medicine spilt straight away.’ [Dâw, Makú; [Aikhenvald 2006](#)]
- Widespread in languages of W Africa, SE Asia, Amazonia, Oceania, New Guinea
- The derivation of SVCs can involve a wide range of mechanisms, e.g. complementation, compounding, coordination, adjunction, or a combination thereof ([Baker & Stewart 2002](#), [Aikhenvald 2006](#), [Cleary-Kemp 2015](#))

2.1 SVCs IN AUSTRONESIAN LANGUAGES

- SVCs in these languages show voice (2,3) or ϕ -agreement (4) morphology across strings of two or more verbs.

(2) Äiwoo [Oceanic]:

- mu-ki-âwââ-mana-(Ø)=kâ* (Actor Voice)
2MIN-IPFV-pull.AV-very-AV=DIST
'You catch a lot (of fish).'
- ki-ââ-mana-i-mu=wâ* (Undergoer Voice)
IPFV-pull.UV-very-UV-2MIN=DIST
'You catch a lot (of fish).'

(3) Kavalan [Formosan; Yeh & Huang 2009]:

- m-li-lizaq-ti-qaniawu m-atiw t<m>aqis* (Actor Voice)
AV-RED~happy-PFV-3SG.NOM AV-go <AV>study
'He is going to study happily.'
- ngid-an-na m-lizaq q<m>an ya baut* (Undergoer Voice)
want-UV-3SG.GEN AV-happy <AV>eat NOM fish
'He wanted to eat the fish happily.'

(4) Amarasi [Timoric]:

- hit ta-reko t-fain t-ok ne*
1PL.IN.NOM 1PL.IN-be.well 1PL.IN-return 1PL.IN-with 3SG.ACC
'We are reconciled with him.'
- In neem na-teef n-ok kai.*
3SG.NOM 3SG.come 3SG-meet 3SG-with 1PL.EX.ACC
'He comes to meet with us.'

2.2 THE THREE PATTERNS OF VERBAL CONCORD

- **TYPE IA:** Äiwoo [Oceanic]; Tsou [Formosan]; Chamorro
All verbs in the SVC match for voice.
- **TYPE IB:** Kavalan, Squliq Atayal [Formosan]
The first verb has "true" voice marking, the rest have default AV.
- **TYPE II:** Amarasi, Uab Meto, Helong, Tetun [Timoric]
All elements in the extended verbal projection match for subject ϕ -agreement.
- **Generalization: "highest-or-all pattern".** Marking either goes on the highest element of a series, or on every single element; there are no other options.

3 TYPE I: SYNTACTIC CONCORD

3.1 ASSUMPTIONS ABOUT PHILIPPINE-TYPE VOICE

- ▶ For concreteness, we follow a model where the different voices are represented as different Voice heads, above vP . Non-actor voices have an EPP feature attracting a given argument to its specifier, rendering it the highest DP and thereby the one accessible to further operations (see e.g. Aldridge 2004, 2008, Erlewine 2018).
 - ▷ This is not crucial for our analysis. If you believe Philippine-type voice is really something else and/or somewhere else, feel free to fill in with your favourite model. What's important for us is that it's higher than vP .
- ▶ Regardless of the mechanisms involved in deriving SVCs, we assume voice marking on verbs to be the interaction of a Voice head with lower v heads (Harley 2013).

3.2 THE INTERACTION/SATISFACTION MODEL OF AGREE

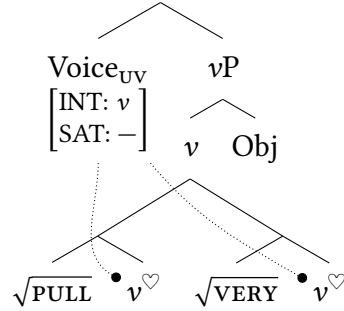
- ▶ Patterns of full voice matching (TYPE IA) and default voice (TYPE IB) can be captured by an interaction/satisfaction (INT/SAT) approach to Agree (Deal 2015, 2021)
- ▶ INT/SAT approach:
 - ▷ There are no unvalued/uninterpretable features. All probes are specified for i) what features they may *interact* with to establish feature copying, and ii) what features may *satisfy* them to end further probing.
 - ▷ *Goal flagging* (Deal 2022a): probes assign certain features to goals indicating that the goal has either interacted with or satisfied the probe via Agree.
 - ▷ Every probe may potentially flag its goal(s) in two ways: one for having interacted, and one for having satisfied.
 - ▷ These flags are invariant: a given probe X will always assign the same flag Y, regardless of the feature values involved.
 - ▷ What goal flagging can't do is gather features from some goal, and then spread these features to other goals (i.e. ✗ downwards valuation)
 - ▷ Probes can not only gather information (i.e. features/values) from their goals, but also change them with some morphosyntactic effect (Chomsky 2001)
- ▶ This model of Agree has been successfully applied to a wide variety of empirical domains, including ϕ -agreement, PCC, information structural movement, negative concord, etc. Especially the “highest-or-all” pattern is recurrent in several of these phenomena. See Deal (2015, 2021, 2022a,b) and references therein .

3.3 SYNTACTIC CONCORD AS AGREE

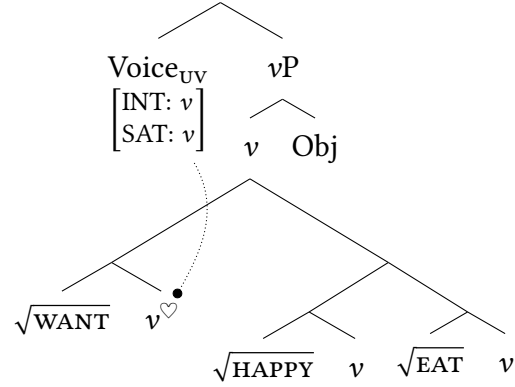
► Two types of syntactic concord: **highest or all**

- ▷ TYPE IA: voice matching as an *insatiable* Voice probe [INT: v ; SAT: $-$]
- ▷ TYPE IB: default voice as a *satiable* Voice probe [INT: v ; SAT: v]

(5) a. **Insatiable probe:** Äiwoo



b. **Satiable probe:** Kavalan



- ▷ TYPE IA, *insatiable probe* (5a): the probe assigns a diacritic feature $\heartsuit$ to every goal it interacts with. Since there is no satisfaction condition, it simply flags all the v 's in its domain.
- ▷ TYPE IB, *satiable probe* (5b): the probe interacts with the first v it finds, and assigns it with a diacritic feature $\heartsuit$. Since it is also satisfied by that v , it doesn't flag any subsequent lower v 's.
- The apparent homophony between default and AV forms in TYPE IB languages can be generalized if AV is simply an absence of Voice features on v .¹
 - ▷ AV would constitute the “unmarked member” of the voice alternation
 - ▷ Cf. singular as the absence of a (privative) [PL] feature; 3rd person as the absence of person features (Benveniste 1971, Harley & Ritter 2002 a.o.); nominative as the absence of case features (Kornfilt & Preminger 2015).
 - ▷ Cf. Chen (2010) on AV~default voice homophony in Mayrinax Atayal.
- In languages with multiple non-default voices (e.g. PV, IV, LV, etc.), the goal-flag would vary depending on the identity of the Voice head. Voice_{UV} assigns $\heartsuit$, Voice_{IV} assigns $\clubsuit$, etc. This is different from co-variant morphological concord (see below).

¹ Our claim that AV is “default” is limited to its morphosyntactic featural make-up. It is known that in several Philippine-type languages, AV is used with less frequency than UV in natural speech (see Liao 2004 for Kavalan and Holmen 2020 for Äiwoo). Consider a parallel situation: in a language like Latin, the nominative is the morphosyntactic default, but there is no expectation that this entails higher frequency than other case forms. In fact, modern Romance languages derive their nouns from the Latin accusative, not the nominative, which is a possible indication of their relative frequency.

3.4 CASE STUDY, TYPE IA: FULL VOICE CONCORD IN ÄIWO

3.4.1 EMPIRICAL PICTURE

- Transitive stems have both AV and UV forms (often with idiosyncratic/suppletive morphology); intransitive stems don't participate in this.
 - ▷ (The two voices also differ in whether ϕ -agreement is realized as a prefix or as a suffix; this is irrelevant here)
- If the first stem of the SVC is in UV, other serialized stems have to take a suffix *-i* (6a); if the first one is AV, no such suffix (6b)

- (6) a. *ki-[ââ-mana-i]-mu=wâ*
 IPFV-pull.UV-very-UV-2MIN=DIST
 ‘You catch a lot (of fish).’
 b. *mu-ki-[âwââ-mana-(Ø)]=kâ*
 2MIN-IPFV-pull.AV-very-AV=DIST
 ‘You catch a lot (of fish).’

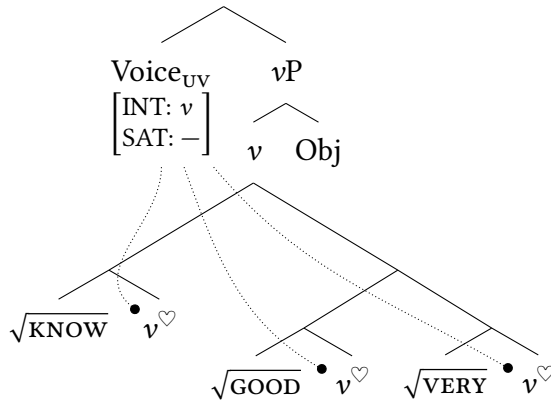
- The “UV voice concord” suffix *-i* appears recursively on every serialized stem:

- (7) a. *bäli enge=ke i-[kää-päko-i]-no*
 side DEM.PROX=PROX ASP-know.UV-good-UV-1MIN
 ‘I know this topic well’
 b. *i-[kää-päko-i-mana-i]-no*
 ASP-know.UV-good-UV-very-UV-1MIN
 ‘I know this very well’
 c. *ki-[eâmole-wâtu-i-päko-i-mana-i]-i ijii=le*
 IPFV-look.UV-COMP-UV-good-UV-very-UV-3AUG 3AUG=PROX
 ‘They have to look after them more properly’

3.4.2 ANALYSIS: DERIVING AND SPELLING OUT CONCORD

- Our approach to voice concord: a probe on Voice_{UV} agrees with every single *v* head in its c-command domain, and flags them with a diacritic (♡)
- ▷ (The specific relative hierarchical position of the three stems within the complex *v* node is purely illustrative, and ultimately unimportant since the probe would interact with all of them anyway.)

(8)



► How do you spell out $\heartsuit$? In a DM world (Halle & Marantz 1993):

- ▷ Assumption: AV is morphosyntactic default, i.e. the absence of a particular Voice feature rather than a feature in and of itself.
- ▷ A large set of transitive stems have their own idiosyncratic VI rules in the presence of $\heartsuit$ vs. without (9). This is independently needed in any account of Äiwoo voice morphology, since there's a lot of unpredictable morphology. One method of formalization is with spans (Svenonius 2012).
- ▷ Many other transitive stems do not have their own idiosyncratic spell-out rule, so $\heartsuit$ is spelled out by itself instead of as part of a span, as *-i* (10).

(9) Some VI rules for idiosyncratic stems:

- | | |
|--|--|
| a. UV: $\langle \sqrt{\text{PULL}}, v^{\heartsuit} \rangle \Leftrightarrow \hat{a}\hat{a}$ | c. UV: $\langle \sqrt{\text{HIT}}, v^{\heartsuit} \rangle \Leftrightarrow \text{togulo}$ |
| AV: $\sqrt{\text{PULL}} \Leftrightarrow \hat{a}\hat{w}\hat{a}\hat{a}$ | AV: $\sqrt{\text{HIT}} \Leftrightarrow \text{togo}$ |
| b. UV: $\langle \sqrt{\text{EAT}}, v^{\heartsuit} \rangle \Leftrightarrow \text{ngä}$ | d. UV: $\langle \sqrt{\text{WEAVE}}, v^{\heartsuit} \rangle \Leftrightarrow \text{vili}$ |
| AV: $\sqrt{\text{EAT}} \Leftrightarrow \text{vängä}$ | AV: $\sqrt{\text{WEAVE}} \Leftrightarrow \text{vei}$ |

(10) Morphological default/elsewhere VI rules:

- $v \Leftrightarrow -i / _{}^{\heartsuit}$ (UV elsewhere form)
 $v \Leftrightarrow \emptyset$ (AV elsewhere form)

► A morpho-phonological argument (Næss 2015):

- ▷ Descriptively, the largest class of alternating stems has a UV form in *-i* (Næss 2015, 2021, Roversi 2019). For example, *epave*, *epavi* 'cook.AV, cook.UV', etc.
- ▷ Although it's not necessarily obviously segmentable, this final /i/ behaves phonologically more like a suffix than like part of a stem.
- ▷ Unstressed /i/ and /u/ are normally devoiced or completely dropped in Äiwoo: *bolevi* 'shore' = [mbo'lev].
- ▷ The final /i/ of (this class of) UV verbs doesn't do this, and it's always pronounced fully: *epavi* 'cook.UV' is pronounced [e'pavi], never *[e'pav].

- If /i/ really is the default spell-out of $\heartsuit$, predictions:
 - ▷ Intransitive predicates are normally never found in the syntactic context of $\heartsuit$ (because they don't participate in the voice alternation)...
 - ▷ So there's no expectation that they should have an idiosyncratic realization in the context of $\heartsuit$...
 - ▷ So if they do end up in the vicinity of $\heartsuit$, we should expect the default realization of $\heartsuit = -i$.
 - ▷ That's exactly what we find in SVCs: intransitive predicates suddenly have a "UV form", and this is formed by adding *-i*, just like the largest morphological class of actually alternating transitive verbs.

3.5 CASE STUDY, TYPE IB: DEFAULT VOICE IN KAVALAN

3.5.1 EMPIRICAL PICTURE

- The initial verb of an SVC can be in AV (11a) or uv (11b) form, while the following verbs are always in AV (or default) form.
 - ▷ The AV marker on verbs can surface as the prefix *m-* or infix $\langle m \rangle$; adjectival and intransitive stems usually take the prefix and transitive stems usually take the infix (Hsieh 2018). The UV marker on verbs is the suffix *-an*.

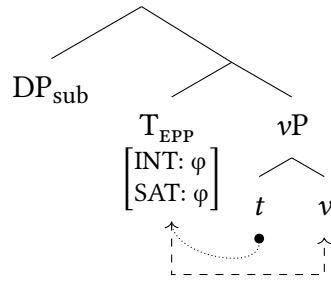
- (11) a. *m-li~lizaq-ti-qaniawu* *m-atiw* *t<m>aqis* [Yeh & Huang 2009]
 AV-RED~happy-PFV-3SG.NOM AV-go $\langle$ AV $\rangle$ study
 'He is going to study happily.'
- b. *ngid-an-na* *m-lizaq* *q<m>an* *ya* *baut*
 want-UV-3SG.GEN AV-happy $\langle$ AV $\rangle$ eat NOM fish
 'He wanted to eat the fish happily.'

- TAM marking, pronominal clitics, and reduplication occur on the initial verb
- In SVCs, the non-final verbs are typically adverbial, adjectival, or modal, while the final verb typically denotes a motion or action.

3.5.2 ANALYSIS: DERIVING AND SPELLING OUT DEFAULT VOICE

- Our approach to default voice:
 - ▷ UV constructions (12): a probe on Voice_{UV} agrees with the first *v* head in its c-command domain, and flags it with the diacritic $\heartsuit$. The probe is satisfied, and no further probing occurs.
 - ▷ AV constructions: there is no probe on Voice_{AV} (or there is a probe similar to Voice_{UV} , but either it does not goal-flag or spell-out is underspecified).

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4.1 CASE STUDY, TYPE II: FULL φ -CONCORD IN AMARASI

4.1.1 EMPIRICAL PICTURE

- All verbs in a SVC string bear full φ -agreement with the shared subject

(15) *hit ta-reko t-fain t-ok ne*
 1PL.IN.NOM 1PL.IN-be.well 1PL.IN-return 1PL.IN-with 3SG.ACC
 ‘We are reconciled with him.’

- One prosodic domain
 - ▷ All non-initial verbs must undergo final CV $\rightarrow$ VC metathesis (e.g. $\sqrt{fani}$ ‘return’ $\rightarrow$ *fain*) unless producing a CCC cluster (e.g. $\sqrt{reko}$ ‘good’ $\rightarrow$ *reok* blocked due to /ktf/)
 - ▷ One domain for primary stress assignment (Mooney 2021, Tan 2021)
- One syn/sem domain
 - ▷ Bracket negation overtly contains + scopes over the entire SVC
 - ▷ Aspectual/Mood markers precede or follow the entire SVC string
- Verb strings which show φ -mismatches are in fact biclausal constructions with pro-dropped subjects, which can reappear (16a)
- Real (semi-)lexical SVCs cannot be interceded (16b):³

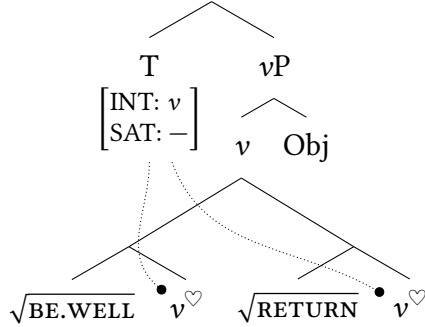
(16) a. *Ho m-tital (mei) neem=ii* [Benu 2014: 86, ex. 8-9]
 2SG 2SG-push (table) 3SG.come=DEM.PROX
 ‘You push (a table) over here.’
 b. *Hai m-heck *{fafi} m-aan {fafi}*
 1PL.EX.NOM 1PL.EX-catch pig 1PL.EX-get pig
 ‘We catch and get a pig.’

³ A small set of prepositional verbs show adjunct-like behaviour in SVCs, allowing adverbs, nominals, bracket negation, and aspectual enclitics to intervene between them and preceding lexical/aspectual verbs. These precise verbs seem to occupy a separate prosodic domain for stress, and are also the only verbs which may show default 3sg agreement rather than concord in an SVC. The fact that they *can* show concord, however, disfavours an analysis in which concord requires all verbs to head-move into a single complex T head.

4.1.2 Φ -CONCORD IS NOT GOAL-FLAGGING

- For ϕ -concord to fully parallel Voice concord on an INT/SAT account, we should assume that T goal-flags v upon entering into whatever syntactic dependency allows for conventional feature-spreading.
- Further paralleling Type I, let us assume this probe to be insatiable:

(17)



- Crucially, ♥ can only be spelled out:
 - ▷ Invariantly, as in Kavalan; or
 - ▷ As conditioned by local allomorphy and/or in portmanteaux/spans with the root, as in much of Äiwoo.
- We do not expect ♥ itself to reflect any of the ϕ -features gained by T, e.g. 1PL.EX.⁴
 - ▷ This key constraint on goal-flagging differentiates it empirically from downwards valuation (Deal 2022a).
 - ▷ Concord that co-varies with a feature copied onto the putative probe therefore cannot involve goal-flagging.
- **Prediction:** concord involving features originating on T could be syntactic and show a highest-or-all pattern.

4.1.3 ANALYSIS: ϕ -CONCORD AS MORPHOLOGICAL

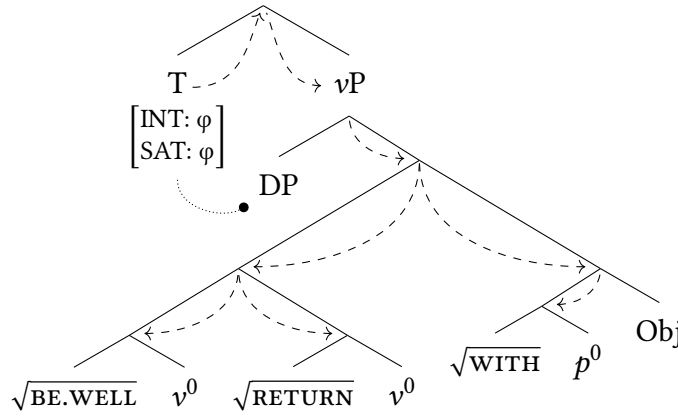
- We adopt a view of ‘morphological’ concord as one that does not strictly rely on an Agree relation, *pace* Baker (2008), Kramer (2014), and Bayırlı (2017) and following Norris (2014, 2018) and Polinsky (2016).
- Key distinguishing characteristic of morphological concord: occurrence on elements occupying a range of distinct syntactic positions (e.g. heads, specifiers, adjuncts) and categories (e.g. D, Quant, Adj, Num, etc.)

⁴ This of course depends on one’s view on the epistemic status of labels – i.e., is T simply shorthand for the collection of features on T, in which case copying ϕ -features from a DP ‘changes’ the label and thus goal-flag of T → e.g. T_{1PL.EX}? This would arguably be functionally equivalent to downwards valuation.

► In a nutshell:

- ▷ T projects the copied ϕ -features up to TP
- ▷ Agr⁰ nodes are post-syntactically inserted onto v^0 , a^0 , p^0 , ... (Kramer 2010)⁵
- ▷ Feature Copying copies ϕ -values from the closest **dominating** projection (i.e. TP) onto the Agr⁰'s

(18)



- By capturing concord through domination/inclusion rather than c-command, other DPs (e.g. internal arguments) do not intervene.
- **Prediction:** concord should occur on various elements in the verbal domain: adjuncts, i.e. prepositions/adverbs
 - ▷ Almost all ‘prepositional’ elements in Amarasi take inflection, and can be independently shown to be adjuncts (Tan 2021).

(19)

3sg Form	Function	Gloss
n-bi	realis locative	(in)to, at
na-'ko	ablative	from
n-oka	comitative	with
n-eu	dative	to, for

- ▷ Many adverbs/modifiers and even quantifiers can take agreement: *(na)-mfaun* ‘many’, *(na)-rek~reko?* ‘well, carefully’, *(na)-mes~mese?* ‘one, alone, completely’

5 OTHER EXISTING THEORIES

- How does our approach to verbal concord compare with existing theories?

⁵ One potential way of unifying syntactic and morphological concord would be to argue that T’s goal-flag on $v - \heartsuit -$ is akin to a diacritic identifying a given head for post-syntactic Agr⁰ node insertion. However, this may struggle to account for concord surfacing on putative non-verbal heads like adverbs and prepositions.

- ▶ A reminder of the key empirical generalization here: **the “highest-or-all” pattern**. That is, morphological marking either goes on the highest element of an SVC, or on every single element; as far as we know, there are no other patterns attested.
- ▶ Some alternative theories:
 - ▷ Upward Agree (Zeijlstra 2012, Bjorkman & Zeijlstra 2014, 2019, Wurmbrand 2014): downwards valuation, where a probe can acquire features from a goal that c-commands it.
 - ▷ Multiple ϕ -probes: every ϕ -agreement marker is a realization of its own head that agrees with the subject.
 - ▷ Multiple Voice projections (Wurmbrand 2013, Wurmbrand & Shimamura 2017): each verb in a series has its own Voice projection, and every voice marker is the realization of its own Voice head.
- ▶ **All three accounts are unable to capture the “highest-or-all” generalization without additional mechanisms.** More specifically:
 - ▷ In the absence of rules governing proper probe distribution, both Upward Agree and multiple ϕ -probe accounts require additional mechanisms to rule out unattested combinations (e.g. highest + lowest only, lowest only).
 - ▷ Multiple ϕ -probes would have to posit an unrestrained distribution of probes, including on putative non-verbs. In addition, depending on the height of these ϕ -probes, one may still require Upward Agree and/or a type of Control dependency between verbs.
 - ▷ Multiple Voice projections would require a type of dependency between Voice heads, as well as the existence of the deficient Voice heads. This is also not compatible with theories where voice is an A' phenomenon.

6 TYPOLOGY AND EMPIRICAL PREDICTIONS

- ▶ It is possible for languages to employ more than one type of verbal concord.
 - ▷ There is no issue of incompatibility with languages having both insatiable and satiable probes, or having probes and morphological feature spreading.
- ▶ Some empirical predictions:
 - ▷ What elements can be marked in verbal concord?
 - ▷ SYNTACTIC → elements that share certain features (e.g. *v*)
 - ▷ MORPHOLOGICAL → all elements, across different categories
 - ▷ Where do features involved in concord originate?
 - ▷ SYNTACTIC → from features on the probe when it is first merged
 - ▷ MORPHOLOGICAL → from features on the probe or those copied back to it

- ▷ What types of features can be involved in concord?
 - ▷ SYNTACTIC → features that originate on probes (e.g. T, Voice)
 - ▷ MORPHOLOGICAL → any features present on the spreading node

(20) Typology of verbal concord:

	Type of concord	What elements receive concord?	Where do concord features originate?	What features are involved?
TYPE IA	SYNTACTIC, <i>insatiable</i>	all (w/ shared feature)	probe	e.g. T, Voice
TYPE IB	SYNTACTIC, <i>satiabile</i>	highest	probe	e.g. T, Voice
TYPE II	MORPHO-LOGICAL	all	probe, or copied features	e.g. T, Voice, ϕ

- Languages in which each type of concord is attested⁶:
 - ▷ TYPE IA: Äiwoo [Oceanic]; Tsou [Formosan]; Chamorro
 - ▷ TYPE IB: Kavalan, Squliq Atayal [Formosan]
 - ▷ TYPE II: Amarasi, Uab Meto, Helong, Tetun [Timoric]

7 CONCLUSION

- We have sketched a typology of verbal concord across Austronesian languages.
- Two main theoretical approaches needed to capture verbal concord patterns:
 - ▷ An interaction/satisfaction approach to Agree with goal-flagging
 - ▷ A post-syntactic approach to morphological spreading
- Open questions:
 - ▷ Are these types of verbal concord observed outside Austronesian languages?
 - ▷ What features are actually attested in syntactic/morphological concord?
 - ▷ Could this have any implications for the analysis of Philippine-type voice?

⁶ Sources of language data include: Äiwoo (fieldwork by Åshild Næss, whom we thank for sharing the data with the third author), Amarasi (fieldwork by second author, Edwards 2020), Chamorro (Wurmbrand & Shimamura 2017), Kavalan (Yeh & Huang 2009, Hsieh 2018), Squliq Atayal (Yeh & Huang 2009), Tsou (Chang 2010, Yeh & Huang 2009), Uab Meto (fieldwork by second author, Benu 2014).

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